

GeoSnap: Land Use Classification and Explainability from Satellite Imagery

Team Information

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1. Introduction

Land-use classification from satellite imagery is an important task in remote sensing. Accurate classification enables monitoring of urban growth, agricultural activity, forest coverage, and environmental changes.

This project uses the EuroSAT dataset containing both RGB imagery and 13-band multispectral Sentinel-2 imagery. Two deep learning models were developed and evaluated:

1. RGB-based CNN
2. Multispectral CNN

The objective is to classify satellite image patches into one of ten land-use categories and analyze model performance.

2. Dataset

The dataset consists of 64×64 satellite image patches belonging to ten classes:

- AnnualCrop
- Forest
- HerbaceousVegetation
- Highway
- Industrial
- Pasture
- PermanentCrop
- Residential
- River
- SeaLake

Two modalities are provided:

- RGB Images (.jpg)
- 13-Band Multispectral Images (.tif)

Training and validation splits were used as provided.

3. Methodology

RGB Model

A convolutional neural network was trained on RGB imagery.

Training Configuration

- Input Size: 64×64×3
 - Optimizer: Adam
 - Loss Function: Cross Entropy Loss
 - Batch Size: 32
 - Hardware: NVIDIA GPU (CUDA)
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Multispectral Model

A custom CNN was developed for 13-band multispectral imagery.

Training Configuration

- Input Size: 64×64×13
- Optimizer: Adam
- Loss Function: Cross Entropy Loss
- Batch Size: 32
- Hardware: NVIDIA GPU (CUDA)

The network processes all thirteen Sentinel-2 bands to exploit additional spectral information unavailable in RGB images.

4. Results

RGB Model Results

Validation Accuracy: 98%

Multispectral Model Results

Validation Accuracy: 91.78%

The multispectral model achieved strong classification performance across all classes.

Insert:

- Confusion Matrix Figure
- Classification Report Figure

5. Explainability Analysis

Confusion Analysis

The confusion matrix reveals several challenging class pairs:

- Pasture ↔ HerbaceousVegetation
- AnnualCrop ↔ PermanentCrop
- Highway ↔ Industrial

These classes share similar spatial structures and spectral characteristics.

Spectral Band Importance

Several Sentinel-2 bands contribute significantly to classification performance:

Near Infrared (B8)

Useful for distinguishing vegetation classes.

Red Edge Bands (B5–B7)

Capture vegetation health and chlorophyll information.

Short-Wave Infrared (B11–B12)

Provide information about moisture and soil composition.

These additional bands explain the superior performance of the multispectral model compared with RGB imagery.

6. Discussion

The results demonstrate the value of multispectral information for remote sensing applications.

While RGB imagery provides visual information, multispectral bands provide additional spectral signatures that help separate visually similar classes.

The model performed particularly well on:

- SeaLake
- Residential
- River
- Forest

The most difficult classes involved agricultural and vegetation categories.

7. Conclusion

Two land-use classification models were developed using RGB and multispectral satellite imagery.

The multispectral model achieved 91.78% validation accuracy and outperformed RGB-based approaches by utilizing information from all thirteen Sentinel-2 bands.

Future work includes:

- Grad-CAM visualization
- SHAP analysis
- Transformer-based architectures
- Multi-modal fusion approaches

References

1. EuroSAT Dataset
2. Sentinel-2 Mission Documentation
3. PyTorch Documentation
4. Scikit-Learn Documentation